

Maximilian Köhlenbeck

Automation Engineering student (work-study) · KUKA & Hochschule Bremen

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Profile	I study Automation Engineering in a work-study program and build systems with AI and robotics. I'm most interested in reinforcement learning, physical AI and systems that interact with the real world. Right now I'm teaching myself how to train models.	
Education	Automation Engineering, work-study program 10/2024 to 12/2027 (expected) Hochschule Bremen (academic phases) and KUKA (practical phases) Abitur (German university entrance qualification) 2024 Gymnasium Vegesack, Bremen	
Projects	Wächter: offline voice and vision assistant 2026 Sees through a camera, listens and talks back, fully offline on a laptop without a GPU. Starts speaking after ~2.6 s (target < 3 s), person detection takes 17 ms per frame. Python · faster-whisper · Piper TTS · local LLM · ONNX/DirectML Baulify: coordinating roadworks (team) since 05/2026 Started at the Smart Region Hackathon Offenburg and developed further afterwards: conflict detection on a map, AI intake from PDFs and free text, approvals via a link without an account. Next.js · TypeScript · PostgreSQL · Drizzle · MapLibre · LLM 3D printer: nozzle offset calibration (with KUKA) ● in progress Optimizing a 3D printer with a nozzle offset calibration I built myself. Tarp editor: wrinkle simulation 2026 Spring-mass simulation of a tarp with a wrinkle index and an assistant that suggests mounting points for a flatter tarp. JavaScript · Canvas Reinforcement learning: CartPole and double pendulum ● in progress PPO agent for CartPole, then a double pendulum on a cart that switches between four poses with a single TQC model. Python · Gymnasium · Stable-Baselines3 (PPO, TQC) · MuJoCo Local dictation (Wispr Flow clone) and n8n automations 2026 Hotkey dictation into any app, fully offline. AI automations for invoices, email triage and price tracking with built-in checks. faster-whisper · n8n · LLM APIs	
Hackathons	Smart Region Hackathon Offenburg 05/2026	
Skills	Reinforcement learning & simulation Local AI & speech Software & web Automation & tooling Hardware Languages	Gymnasium, Stable-Baselines3 (PPO, SAC, TQC), MuJoCo, spring-mass simulation faster-whisper, Piper TTS, Ollama, ONNX / DirectML, LLMs via OpenRouter Python, TypeScript, JavaScript, Next.js, Astro, PostgreSQL, Drizzle, MapLibre n8n, Claude Code, Codex, Git 3D printing German (native), English (C1)